

UNIT-2

LEETCODEEXERCISE

1.Find Users with Valid E-Mails

Problem Summary

You are given a table named Users.

Column Name	Type
user_id	int
Name	Varchar
Mail	Varchar(50)

- user_id is the primary key.
- Each row contains the ID and name (which is an e-mail address) of a user.

You need to find users whose e-mail addresses are valid

Valid E-mail Rules

A valid e-mail:

1. Starts with a letter (a-z or A-Z)
2. Followed by:
 - o letters (a-z, A-Z)
 - o digits (0-9)
 - o underscore (_)
 - o period (.)
 - o hyphen (-)
3. Ends [with:@leetcode.com](https://leetcode.com)

Task

Write an SQL query to find all users having valid e-mail addresses.

Return the result table in any order.

MySQL Solution

MySQL   Auto

```
1 # Write your MySQL query statement below
2 SELECT user_id, name, mail
3 FROM Users
4 WHERE REGEXP_LIKE(
5     mail,
6     '^[A-Za-z][A-Za-z0-9._-]*@leetcode\\.com$',
7     'c'
8 );
```

Output

user_id	name	mail
1	Winston	winston@leetcode.com
3	Annabelle	bella@leetcode.com
4	Sally	sally.come@leetcode.com

Expected

user_id	name	mail
1	Winston	winston@leetcode.com
3	Annabelle	bella@leetcode.com
4	Sally	sally.come@leetcode.com

2. Patients With a Condition

Problem Summary

You are given a table named Patients.

Column Name	Type
patient_id	int
patient_Name	Varchar
conditions	Varchar

- patient_id is the primary key.

Important Rule

A patient is considered valid if:

- DIAB1 appears at the beginning of the string, or
- DIAB1 appears after a space.

Task

Write an SQL query to report the patient ID, patient name, and conditions of patients who have Type I Diabetes (DIAB1).

Return the result table in any order.

MySQL Solution

MySQL   Auto

```

1 # Write your MySQL query statement below
2 SELECT
3     patient_id,
4     patient_name,
5     conditions
6 FROM Patients
7 WHERE conditions LIKE 'DIAB1%'
8     OR conditions LIKE '% DIAB1%';

```

Output

patient_id	patient_name	conditions
3	Bob	DIAB100 MYOP
4	George	ACNE DIAB100

Expected

patient_id	patient_name	conditions
3	Bob	DIAB100 MYOP
4	George	ACNE DIAB100

3.Second Highest Salary

Problem Summary

You are given an Employee table.

Column Name	Type
id	int
salary	int

- id is the primary key.
- Each row contains the salary of an employee.
- Salaries may contain duplicates.

Your task is to find the second highest distinct salary.

If there is no second highest salary, return NULL.

Task

Write an SQL query to report the second highest distinct salary from the Employee table.

SQL Solution (Using LIMIT and OFFSET)

MySQL   Auto

```
1 # Write your MySQL query statement below
2 SELECT
3 (
4     SELECT DISTINCT salary
5     FROM Employee
6     ORDER BY salary DESC
7     LIMIT 1 OFFSET 1
8 ) AS SecondHighestSalary;
```

Output

```
| SecondHighestSalary |
| ----- |
| 200 |
```

Expected

```
| SecondHighestSalary |
| ----- |
| 200 |
```

4.Customer Placing the Largest Number of Orders

Problem Summary

You are given an Orders table.

Column Name	Type
order_number	int
customer_number	int

- order_number is unique.
- Each row represents one order placed by a customer.
- A customer can place multiple orders.

Your task is to find the customer who placed the largest number of orders.

The problem guarantees that:

There is exactly one customer who has placed the maximum number of orders.

Task

Write an SQL query to find the customer_number of the customer who has placed the largest number of orders.

Return the result table in any order.

MySQL Solution (GROUP BY + ORDER BY + LIMIT)

MySQL   Auto

```
1 # Write your MySQL query statement below
2 SELECT customer_number
3 FROM Orders
4 GROUP BY customer_number
5 ORDER BY COUNT(*) DESC
6 LIMIT 1;
```

Output

```
| customer_number |  
| ----- |  
| 3 |
```

Expected

```
| customer_number |  
| ----- |  
| 3 |
```

5.Product Sales Analysis III

Problem Summary

You are given two tables, but only the Sales table is needed for this problem.

Sales Table

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

- (sale_id, year) is the primary key.
- Each row represents a sale of a product in a particular year.

Task

Write an SQL query to report:

- product_id
- first_year
- quantity
- price

for the first year in which each product was sold.

Return the result table in any order.

MySQL Solution (GROUP BY + JOIN)

MySQL   Auto

```
1 # Write your MySQL query statement below
2 SELECT
3     product_id,
4     year first_year,
5     quantity,
6     price
7 FROM Sales
8 WHERE (product_id, year) IN (
9     SELECT product_id, MIN(year)
10    FROM Sales
11   GROUP BY product_id
12 );
```

Output

product_id	first_year	quantity	price
100	2008	10	5000
200	2011	15	9000

Expected

product_id	first_year	quantity	price
100	2008	10	5000
200	2011	15	9000